

Robust Controller Synthesis

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Abstract—

*Index Terms—*PDEs, Robust Control, LMIs;

I. INTRODUCTION

This paper uses PN PI operators inspired by the works [2], [5], [14]. Then it IQC extends the results from [1], [4], [11]. [3] also covers robust control using dual iq

STRUCTURE OF THE PAPER

The structure of the paper will be organized as follows:

- **Preliminaries:** This section introduces the necessary notation, concepts related to PI and PIETOOLS, and the IQC result that will be central to the subsequent developments in the paper.
- **Scaled IO Result:** In this section, we define the scalings associated with both the primal and dual systems. These scalings play a critical role in the connection between the primal and dual formulations. Then we will use these scalings in a scaled IO duality result.
- **Connection to IQC:** We then connect the scaled duality result to IQC. Consider the following hard IQC condition:

$$\int_0^T \left\langle \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}, \Pi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix} \right\rangle dt \geq 0$$

where the dynamic multiplier $\Pi(j\omega)$ can be decomposed into a dynamic part $\Psi(j\omega)$ and a static part $\mathcal{V}$. The IQC condition then becomes:

$$\int_0^T \left\langle \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}, \mathcal{V} \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix} \right\rangle dt \geq 0$$

We propose a further decomposition of $\mathcal{V}$ as:

$$\mathcal{V} = \mathcal{S}^* \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \mathcal{S}$$

where $\mathcal{S}$ represents the scaling operations applied to the output of $\Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}$. Assuming Ψ admits a PIE system, we augment the nominal system as $z_\Delta = Gw_\Delta$. Rewriting the IQC condition with this decomposition gives:

$$\int_0^T \left\langle \mathcal{S} \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}, \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix} \mathcal{S} \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix} \right\rangle dt$$

This formulation reduces to analyzing the open-loop system $\mathcal{S} \Psi \begin{bmatrix} z_\Delta(t) \\ w_\Delta(t) \end{bmatrix}$. After establishing the necessary

assumptions and conditions, we apply the scaled duality result to the augmented system with corresponding scalings.

- **Controller Synthesis:** In this section, we propose Linear PI Inequalities (LPIs) for robustly synthesizing controller gains. If time allows, we will extend this to include output feedback. Additionally, we may explore using the DK iteration method to solve for suboptimal solutions when combining dynamic multipliers with output feedback.
- **Examples and Validation:** Finally, we explore practical applications, focusing on performance shaping using weighting filters. The results will be validated through Frequency Response Function (FRF) measurements.

II. PRELIMINARIES

A. Notation

$L_2^n[a, b]$ denotes the set of $\mathbb{R}^n$ -valued, Lebesgue square-integrable equivalence classes of functions on the spatial domain $[a, b] \subset \mathbb{R}$. The space $\mathbb{R}L_2^n[a, b]$ denotes the Cartesian product $\mathbb{R}^{n_1} \times L_2^{n_2}[a, b]$, where $\mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$, with inner product

$$\left\langle \begin{bmatrix} x_1 \\ \mathbf{x}_2 \end{bmatrix}, \begin{bmatrix} y_1 \\ \mathbf{y}_2 \end{bmatrix} \right\rangle_{\mathbb{R}L_2} := x_1^\top y_1 + \langle \mathbf{x}_2, \mathbf{y}_2 \rangle_{L_2}.$$

When clear from context, we occasionally omit the subscript and simply write $\langle \cdot, \cdot \rangle := \langle \cdot, \cdot \rangle_{\mathbb{R}L_2}$. Since the only spatial domain considered here is $[a, b]$, we typically omit the domain and write L_2^n or $\mathbb{R}L_2^n$, further omitting dimensions when clear from context. For a normed space X and Banach space Y , $\mathcal{L}(X, Y)$ denotes the Banach space of bounded linear operators from X to Y with induced norm

$$\|\mathcal{P}\|_{\mathcal{L}(X, Y)} = \sup_{\|\mathbf{x}\|=1} \|\mathcal{P}\mathbf{x}\|_Y,$$

and $\mathcal{L}(X) := \mathcal{L}(X, X)$. Our notational convention is to write functions in **bold** (e.g., $\mathbf{x}$, indicating $\mathbf{x} \in \mathbb{R}L_2$) and operators in calligraphic font (e.g., $\mathcal{P} \in \mathcal{L}(\mathbb{R}L_2)$). For a Hilbert space X and operator $\mathcal{A} \in \mathcal{L}(X)$, the adjoint $\mathcal{A}^*$ is defined by

$$\langle \mathbf{x}, \mathcal{A}\mathbf{y} \rangle_X = \langle \mathcal{A}^* \mathbf{x}, \mathbf{y} \rangle_X, \quad \forall \mathbf{x}, \mathbf{y} \in X.$$

The space $L_2^n[0, \infty)$ denotes $\mathbb{R}^n$ -valued square-integrable signals defined on the temporal domain $[0, \infty)$, so that $x(t) \in \mathbb{R}^n$. Similarly, $\mathbb{R}L_2^n[0, \infty)$ denotes $\mathbb{R}L_2$ -valued square-integrable temporal signals. For suitably differentiable $\mathbf{x} \in \mathbb{R}L_2^n[0, \infty)$, we write $\dot{\mathbf{x}} := \frac{\partial \mathbf{x}}{\partial t}$. For a Hilbert space $\mathbf{H}$, we denote by $\mathbf{L}_\mathbf{H}$ the space of square-integrable functions $u : [0, \infty) \rightarrow \mathbf{H}$ with inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{L}_\mathbf{H}} = \int_0^\infty \langle \mathbf{u}(t), \mathbf{v}(t) \rangle_{\mathbf{H}} dt.$$

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We define the extended space $\mathbf{L}_{e,\mathbf{H}}$ to consist of functions $u : [0, \infty) \rightarrow \mathbf{H}$ such that

$$\int_0^T \|u(t)\|_{\mathbf{H}}^2 dt < \infty \quad \forall T \geq 0.$$

We use $\mathbf{L}_{e,[a,b]}$ and $\mathbf{L}_{[a,b]}$ to denote $\mathbf{L}_{e,\mathbf{H}}$ and $\mathbf{L}_{\mathbf{H}}$ with $\mathbf{H} = \mathbb{R}L_2[a, b]$. For any Hilbert space $\mathbf{H}$, the truncation operator $p_\tau : \mathbf{L}_{e,\mathbf{H}} \rightarrow \mathbf{L}_{e,\mathbf{H}}$ is defined for $\mathbf{y} \in \mathbf{L}_{e,\mathbf{H}}$ by

$$(p_\tau \mathbf{y})(t) = \begin{cases} \mathbf{y}(t), & 0 \leq t \leq \tau, \\ 0, & \text{otherwise.} \end{cases}$$

such that $(p_\tau \mathbf{y})(t) \in \mathbf{L}_{\mathbf{H}}$ for all $\tau \geq 0$.

Definition 1. Let $\mathbf{H}$ and $\mathbf{G}$ be Hilbert spaces. For all $\tau \geq 0$, an operator $G : \mathbf{L}_{\mathbf{H}} \rightarrow \mathbf{L}_{\mathbf{G}}$ is:

- **Causal** if $p_\tau G = p_\tau G p_\tau$.
- **Bounded on $\mathbf{L}_{\mathbf{H}}$** if $\|G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|\mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$ for some $C \geq 0$ and all $\mathbf{v} \in \mathbf{L}_{\mathbf{H}}$.
- **Bounded on $\mathbf{L}_{e,\mathbf{H}}$** if $\|p_\tau G\mathbf{v}\|_{\mathbf{L}_{\mathbf{G}}} \leq C\|p_\tau \mathbf{v}\|_{\mathbf{L}_{\mathbf{H}}}$ for some $C \geq 0$ and all $\mathbf{v} \in \mathbf{L}_{e,\mathbf{H}}$.

Finally, for a given $\mathcal{K} \in \mathcal{L}(\mathbb{R}L_2^n, \mathbb{R}L_2^m)$, we define the operator $\mathcal{M}_{\mathcal{K}} : \mathbf{L}_{e,[a,b]}^n \rightarrow \mathbf{L}_{e,[a,b]}^m$ to compactly represent a multiplication operator. For $\mathbf{u} = [u_1 \cdots u_N]^\top$ with $\mathbf{u} \in \mathbb{R}L_2$, its action at time t is given by the block structure of $\mathcal{K}_{ij}$:

$$(\mathcal{M}_{\mathcal{K}}\mathbf{u})(t) := \begin{bmatrix} \mathcal{K}_{11} & \cdots & \mathcal{K}_{1n} \\ \vdots & \ddots & \vdots \\ \mathcal{K}_{m1} & \cdots & \mathcal{K}_{mn} \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 \\ \vdots \\ \mathbf{u}_n \end{bmatrix} (t). \quad (1)$$

B. Partial Integral Equations (PIEs)

Partial Integral operators, also known as PI operators, are a class of bounded linear integral operators that are defined jointly on a vector space and Hilbert space.

Definition 2. We say $\mathcal{P} \in \Pi_4^{m \times n} \subset \mathcal{L}(\mathbb{R}L_2^n, \mathbb{R}L_2^m)$, denoted by $\Pi \begin{bmatrix} P & Q_1 \\ Q_2 & \{R\} \end{bmatrix}$, if there exist a matrix P and matrix polynomials Q_1, Q_2, R_0, R_1 , and R_2 (of compatible dimension) such that,

$$\left[\left(\Pi \begin{bmatrix} P & Q_1 \\ Q_2 & \{R\} \end{bmatrix} \right) \begin{bmatrix} x \\ \mathbf{x} \end{bmatrix} \right] (s) := \begin{bmatrix} Px + \int_a^b Q_1(s) \mathbf{x}(s) ds \\ Q_2(s)x + \mathcal{R}\mathbf{x}(s) \end{bmatrix},$$

$$(\mathcal{R}\mathbf{x})(s) = R_0(s)\mathbf{x}(s) + \int_a^s R_1(s, \theta)\mathbf{x}(\theta) d\theta + \int_s^b R_2(s, \theta)\mathbf{x}(\theta) d\theta.$$

The vector space of PI operators, given compatible dimensions, are closed under composition, addition, adjoint and concatenation – implying that Π_4 is a composition algebra [10].

Definition 3. (LPIs) For given PI operators $\{\mathcal{E}_{i,j}, \mathcal{F}_{i,j}, \mathcal{G}_i\}$ and a convex linear functional $\mathcal{L}(\cdot)$, a linear PI inequality is a convex optimization of the following form

$$\begin{aligned} & \min_{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}} \mathcal{L}(\{P_i, Q_{1i}, Q_{2i}, R_{0i}, R_{1i}, R_{2i}\}) \\ & \text{s.t.} \quad \sum_{j=1}^K \mathcal{E}_{ij}^* \Pi \begin{bmatrix} P_i & Q_{1i} \\ Q_{2i} & \{R_i\} \end{bmatrix} \mathcal{F}_{ij} + \mathcal{G}_i \succeq 0 \end{aligned} \quad (2)$$

Only LMIs are required to solve an optimization problem involving LPIs. These definitions are based on [7]

Definition 4 (PIE System). Given the PIE system $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\}$ with PI operators $\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}$, and $\mathcal{D}$ of compatible dimensions, and signals $w(t) \in \mathbb{R}L_2^{n_w}[a, b]$, we say $\mathbf{x}(t) \in \mathbb{R}L_2[a, b]^{n_x}$ and $z(t) \in \mathbb{R}L_2^{n_z}[a, b]$ satisfy the PIE with initial condition $\mathbf{x}_0$ if

$$\begin{bmatrix} \partial_t(\mathcal{T}\mathbf{x})(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{C} & \mathcal{D} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ w(t) \end{bmatrix} \quad (3)$$

for all $t \geq 0$ and $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$. Additionally, for signals $w(t) \in \mathbb{R}L_2[a, b]^{n_w}$ of compatible dimensions, we say $\underline{\mathbf{x}}(t) \in \mathbb{R}L_2[a, b]^{n_x}$ and $\underline{z}(t) \in L_2[a, b]^{n_w}$ satisfy the dual PIE, $G^\top$ with initial condition $\underline{\mathbf{x}}_0$ if

$$\begin{bmatrix} \mathcal{T}^* \underline{\dot{\mathbf{x}}}(t) \\ \underline{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & \mathcal{C}^* \\ \mathcal{B}^* & \mathcal{D}^* \end{bmatrix} \begin{bmatrix} \underline{\mathbf{x}}(t) \\ \underline{w}(t) \end{bmatrix}, \quad (4)$$

for all $t \geq 0$ and $\mathcal{T}^* \underline{\mathbf{x}}(0) = \mathcal{T}^* \underline{\mathbf{x}}_0$.

Definition 5 (Generalization of RH_∞). Let G be a PIE system as in Definition 4. We say G is internally stable and bounded with decay rate $\alpha > 0$, $M > 0$, and L_2 -gain $\gamma > 0$ if

- The homogeneous part $\partial_t(\mathcal{T}\mathbf{x})(t) = \mathcal{A}\mathbf{x}(t)$, with $\mathcal{T}\mathbf{x}(0) = \mathcal{T}\mathbf{x}_0$ is exponentially stable such that, for every admissible initial condition $\mathbf{x}_0 \in \mathbb{R}L_2$

$$\|\mathcal{T}\mathbf{x}(t)\|_{\mathbf{L}_{[a,b]}} \leq M\|\mathcal{T}\mathbf{x}_0\|_{\mathbf{L}_{[a,b]}} e^{-\alpha t}, \quad \forall t \geq 0.$$

- With zero initial condition $\mathbf{x}(0) = 0$, then for any input w and output z ,

$$\|p_\tau z\|_{\mathbf{L}_{[a,b]}} \leq \gamma \|p_\tau w\|_{\mathbf{L}_{[a,b]}}, \quad \forall \tau \geq 0, \quad \forall w \in \mathbf{L}_{e,[a,b]}^{n_w}.$$

Then $G \in H_\infty$. Clearly, $RH_\infty \subset H_\infty$ [12]. Also $G^\top \in H_\infty$ with the same decay rate, L_2 -gain, and M [7].

Definition 6 (Linear Fractional Transformation (LFT)). Let G have two input-output pairs, and let H be a compatible causal system or operator with either two ports or one port. Write

$$\begin{bmatrix} y_{11} \\ y_{12} \end{bmatrix} = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix} \begin{bmatrix} u_{11} \\ u_{12} \end{bmatrix}.$$

The Redheffer star products [6] are the following well-posed interconnections.

- If H has two ports,

$$\begin{bmatrix} y_{21} \\ y_{22} \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} u_{21} \\ u_{22} \end{bmatrix},$$

then

$$\begin{aligned} G \star H : \begin{bmatrix} u_{12} \\ u_{21} \end{bmatrix} &\mapsto \begin{bmatrix} y_{12} \\ y_{21} \end{bmatrix}, & u_{11} = y_{22}, & u_{22} = y_{11}, \\ H \star G : \begin{bmatrix} u_{11} \\ u_{22} \end{bmatrix} &\mapsto \begin{bmatrix} y_{11} \\ y_{22} \end{bmatrix}, & u_{21} = y_{12}, & u_{12} = y_{21}. \end{aligned}$$

- If H has one port, $y_H = Hu_H$, then

$$G \star H : u_{12} \mapsto y_{12}, \quad u_{11} = y_H, \quad u_H = y_{11},$$

$$H \star G : u_{11} \mapsto y_{11}, \quad u_H = y_{12}, \quad u_{12} = y_H.$$

C. PIETOOLS

PIETOOLS is a MATLAB toolbox used for declaration, manipulation and solving LPIs [9]. For learning how to use PIETOOLS and its functionalities visit PIETOOLS homepage.

III. IQC-BASED FRAMEWORK FOR ROBUST STABILITY AND PERFORMANCE ANALYSIS

This section develops a comprehensive framework for analyzing stability and performance of interconnected systems in a Hilbert space, utilizing IQCs as the primary analytical tool. We begin by establishing the mathematical foundation for system interconnections and formalizing the key concepts of stability and performance. The IQC methodology is then systematically developed to characterize system uncertainties and derive rigorous conditions that guarantee robust stability and performance. A central contribution of this work lies in bridging IQC theory with μ -analysis for infinite dimensional systems, creating a framework for quantifying their robust stability and performance when those systems are subjected to structured uncertainties.

A. Description of uncertain systems in a Hilbert-space

We start by denoting the uncertainty operators as Δ that belongs to a structured set $\mathbf{\Delta}$ which satisfies the following fundamental property:

Assumption 1. $\Delta \in \mathbf{\Delta}$ implies that $\tau\Delta \in \mathbf{\Delta}$ for all $\tau \in [0, 1]$.

That is, the uncertainty set is closed under scaling by factors in $[0, 1]$, which is the standard homotopy assumption used in IQC theory. We consider operators that map external inputs to external outputs through a feedback interconnection between a linear (infinite dimensional) operator P and Δ . This interconnection structure is formally defined as follows

Definition 7. Given operators $P_\Delta : \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}}$, $P_{\Delta p} : \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}}$, $P_{p\Delta} : \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}}$, $P_p : \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}}$, and $\Delta : \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}}$, define

$$\begin{bmatrix} z_\Delta \\ z_p \end{bmatrix} = \begin{bmatrix} P_\Delta & P_{\Delta p} \\ P_{p\Delta} & P_p \end{bmatrix} \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix}, \quad w_\Delta = \Delta z_\Delta. \quad (5)$$

for input $w_p \in \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$ and output $z_p \in \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}}$, there exist internal signals $z_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}}$ and $w_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$ satisfying the interconnected system $z_p = (P \star \Delta)w_p$.

The interconnection structure described above captures the feedback interaction between the nominal system P and the uncertainty Δ , where w_p and z_p represent external inputs and performance outputs, respectively, while w_Δ and z_Δ form the internal feedback loop through the uncertainty. Before analyzing stability and performance, we must first ensure the well-posedness of the interconnection, such that solutions exist, are unique, and depend causally on inputs.

Definition 8. We say that the uncertain interconnection $P \star \Delta$ is **well-posed** if:

- **Existence and Uniqueness:** For any $w \in \mathbf{L}_{e,[a,b]}$, there exist unique $z \in \mathbf{L}_{e,[a,b]}$, $z_\Delta \in \mathbf{L}_{e,[a,b]}$ and $w_\Delta \in \mathbf{L}_{e,[a,b]}$ such that, z , w , z_Δ , w_Δ satisfy the interconnection.
- **Causality:** If z_Δ , w_Δ and $\hat{z}_\Delta$, $\hat{w}_\Delta$ satisfy the interconnection, then $p_\tau(z_\Delta - \hat{z}_\Delta) = 0$ and $p_\tau(w_\Delta - \hat{w}_\Delta) = 0$, for all $\tau \geq 0$.

Having established the class of uncertain systems and their well-posedness, we now develop a framework to guarantee robust stability and performance for the class of systems defined in Definition 7.

B. Robust stability and performance

Traditional control system design often treats stability and performance shaping as separate objectives with distinct constraints. However, Integral Quadratic Constraints (IQCs) provide a unified framework where these trade-offs can be captured within a single constraint [11]. We now define what it means for the system to remain stable despite uncertainties. Robust stability requires that the system is well-posed for all admissible uncertainties and that the input-output map remains bounded. This involves removal of the performance channels in (5), and introducing auxiliary signals z_0 and w_0 .

Definition 9. The interconnection

$$z_\Delta = P_\Delta w_\Delta + z_0, \quad w_\Delta = \Delta z_\Delta + w_0,$$

with $z_0, w_0 \in \mathbf{L}_{e,[a,b]}$, is said to be **robustly stable** if:

- 1) The uncertain system is well-posed for all $\Delta \in \mathbf{\Delta}$.
- 2) The map from $\begin{bmatrix} z_0 \\ w_0 \end{bmatrix}$ to $\begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}$ is bounded on $\mathbf{L}_{[a,b]}$ for all $\Delta \in \mathbf{\Delta}$.

Within the IQC framework, the stability of an interconnection $P \star \Delta$ is certified by verifying that the uncertainty operator Δ satisfies a quadratic constraint defined by an appropriate multiplier. This leads to the following formal definition

Definition 10. An uncertainty operator $\Delta : \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{z_\Delta}}$ is said to satisfy the IQC defined by the operator $\Psi \in \mathbf{L}_{e,[a,b]}^{n_{w_\Delta} + n_{z_\Delta}} \rightarrow \mathbf{L}_{e,[a,b]}^{n_{w_\Delta} + n_{z_\Delta}}$ multiplier $\mathcal{M}_\Psi \in \mathcal{L}(\mathbf{L}_{[a,b]})$, parameterized by $\mathcal{V}_\Delta = \mathcal{V}_\Delta^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$ if the inequality holds for all $z_\Delta \in \mathbf{L}_{e,[a,b]}$ and all $T \geq 0$,

$$\left\langle p_T \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \geq 0. \quad (6)$$

Moreover, let $\mathcal{V}_\Delta \{ \mathcal{Q}_\Delta, \mathcal{S}_\Delta, \mathcal{R}_\Delta \}$ be parameterized as,

$$\mathcal{V}_\Delta = \begin{bmatrix} \mathcal{Q}_\Delta & \mathcal{S}_\Delta \\ \mathcal{S}_\Delta^* & \mathcal{R}_\Delta \end{bmatrix}$$

Beyond stability, we are interested in ensuring that the system meets certain performance specifications in the presence of uncertainty. Robust performance extends the notion of robust stability by additionally requiring that a performance metric is satisfied.

Definition 11. Robust performance is achieved if the interconnection $P \star \Delta$ is **robustly stable** for some performance multiplier $\mathcal{M}_{\mathcal{V}_p} \in \mathcal{L}(\mathbf{L}_{[a,b]})$, parameterized by $\mathcal{V}_p = \mathcal{V}_p^* \in \mathcal{L}(\mathbb{R}L_2, \mathbb{R}L_2)$ with $\mathcal{Q}_p \succeq 0$ such that, for all $w_p \in \mathbf{L}_{e,[a,b]}$, all $T \geq 0$, and some sufficiently small $\epsilon > 0$,

$$\left\langle p_T \begin{bmatrix} z_p \\ w_p \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \begin{bmatrix} z_p \\ w_p \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_p\|_{\mathbf{L}_{[a,b]}}^2$$

Moreover, let $\mathcal{V}_p\{\mathcal{Q}_p, \mathcal{S}_p, \mathcal{R}_p\}$ be parameterized as,

$$\mathcal{V}_p = \begin{bmatrix} \mathcal{Q}_p & \mathcal{S}_p \\ \mathcal{S}_p^* & \mathcal{R}_p \end{bmatrix}$$

In this framework, the multipliers $\mathcal{M}_{\mathcal{V}}$ and $\mathcal{M}_{\hat{\mathcal{V}}}$ serve distinct but complementary roles. The multiplier $\mathcal{M}_{\mathcal{V}}$ characterizes the behavior of uncertainties, nonlinearities, and other perturbations captured by the operator Δ , ensuring that the IQC is satisfied for the class of admissible disturbances. In contrast, the multiplier $\mathcal{M}_{\hat{\mathcal{V}}}$ encodes performance requirements for the nominal system, such as L_2 -gain bounds, passivity conditions, or other desired input-output properties. Together, these multipliers enable simultaneous verification of robust stability against perturbations and achievement of specified performance objectives.

C. Theorem for joint robust stability and performance

We now present the main theoretical result that provides conditions for simultaneous robust stability and performance verification.

Theorem 1 (Robust Stability and Performance). Let $P, \Delta \in \mathbf{A}$ be given, and there exists $\Psi, \mathcal{M}_{\mathcal{V}_\Delta}$, such that,

- The interconnection $P \star \Delta$ is well-posed for all $\Delta \in \mathbf{A}$
- The IQC in Definition 10 is satisfied for all $\Delta \in \mathbf{A}$.

Then the interconnection $P \star \Delta$ is **robustly stable** and **robust performance** is achieved with respect to some $\mathcal{M}_{\mathcal{V}_p}$, in Definition 11, with $\mathcal{Q}_p \succeq 0$, if,

$$\left\langle p_T \hat{P} \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix}, p_T \begin{bmatrix} \mathcal{M}_{\mathcal{V}_\Delta} & 0 \\ 0 & \mathcal{M}_{\mathcal{V}_p} \end{bmatrix} \hat{P} \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon (\|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 + \|p_T w_p\|_{\mathbf{L}_{[a,b]}}^2) \quad (7)$$

for all $w_\Delta \in \mathbf{L}_{e,[a,b]}^{n_{w_\Delta}}$, $w_p \in \mathbf{L}_{e,[a,b]}^{n_{w_p}}$ and $T \geq 0$, where $\hat{P} :=$

$$\begin{bmatrix} \Psi_\Delta & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} P_\Delta & P_{\Delta p} \\ I & 0 \\ \bar{P}_{p\Delta} & \bar{P}_p \\ 0 & I \end{bmatrix} \text{ according to (5).}$$

Proof. See Appendix A.1 ■

Theorem 1 establishes a framework for deriving Linear PI Inequalities (LPIs), which are used in assessing the robust performance of a system.

IV. DUALITY UNDER DYNAMICAL FILTERING

Theorem 2 (Dual L_2 -gain [7, Thm. 9]). For a PIE system with corresponding PI operators $G\{\mathcal{T}, \mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}\} \in H_\infty$, the following statements are equivalent.

- 1) For any $w \in \mathbf{L}_{e,[a,b]}^{n_w}$, and $\mathbf{x}(0) = 0$, G satisfies $\|P_\tau z\| \leq \|P_\tau w\|$.
- 2) For any $\underline{w} \in \mathbf{L}_{e,[a,b]}^{n_z}$, and $\underline{\mathbf{x}}(0) = 0$, $G^\top$ satisfies $\|P_\tau \underline{z}\| \leq \|P_\tau \underline{w}\|$.

Definition 12 (Primal and Dual Dynamic Filters). Let $\Psi = \begin{bmatrix} \Psi_z & \Psi_{zw} \\ \Psi_{wz} & \Psi_w \end{bmatrix}$ satisfy $\Psi, \Psi^{-1} \in H_\infty$. Let $\hat{\mathcal{J}} := \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix}$. The dual filter associated with Ψ is defined by the dual operation

$$\mathbf{D}(\Psi) := (\hat{\mathcal{J}}^* \Psi^\top \hat{\mathcal{J}})^{-1} = \begin{bmatrix} \Psi_w^\top & -\Psi_{zw}^\top \\ -\Psi_{wz}^\top & \Psi_z^\top \end{bmatrix}^{-1}. \quad (8)$$

Note $\mathbf{D}(\Psi) \in H_\infty$. Since $\hat{\mathcal{J}}^* = \hat{\mathcal{J}}^{-1}$, $\hat{\mathcal{J}}^* = -\hat{\mathcal{J}}$, $(\Psi^\top)^\top = \Psi$, and $(\Psi^{-1})^{-1} = \Psi$, this operation is an involution:

$$\mathbf{D}(\mathbf{D}(\Psi)) = \Psi. \quad (9)$$

The filter Ψ admits the state-space realization

$$\begin{bmatrix} \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi & \mathcal{B}_\Psi^z & \mathcal{B}_\Psi^w \\ \mathcal{C}_\Psi^z & \mathcal{D}_\Psi^z & \mathcal{D}_\Psi^{zw} \\ \mathcal{C}_\Psi^w & \mathcal{D}_\Psi^{wz} & \mathcal{D}_\Psi^w \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ z(t) \\ w(t) \end{bmatrix}, \quad (10)$$

where the entries are PI operators of appropriate dimension. The dual filter $\mathbf{D}(\Psi)$ is equivalently represented through the inverse relation $\mathbf{D}(\Psi)^{-1} = \hat{\mathcal{J}}^* \Psi^\top \hat{\mathcal{J}}$, which has the state-space realization

$$\begin{bmatrix} \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \underline{z}(t) \\ \underline{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi^* & -\mathcal{C}_\Psi^{w*} & \mathcal{C}_\Psi^{z*} \\ -\mathcal{B}_\Psi^{w*} & \mathcal{D}_\Psi^{w*} & -\mathcal{D}_\Psi^{zw*} \\ \mathcal{B}_\Psi^{z*} & -\mathcal{D}_\Psi^{wz*} & \mathcal{D}_\Psi^{z*} \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ \underline{z}(t) \\ \underline{w}(t) \end{bmatrix}. \quad (11)$$

The filtered primal and dual systems are obtained from

$$\Psi \begin{bmatrix} z(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix}, \quad \mathbf{D}(\Psi) \begin{bmatrix} \underline{z}(t) \\ \underline{w}(t) \end{bmatrix} = \begin{bmatrix} \tilde{\underline{z}}(t) \\ \tilde{\underline{w}}(t) \end{bmatrix}. \quad (12)$$

Adopted from [14]

Definition 13. (Potapov–Ginzburg Transform) Given Ψ as in Definition 12, the Potapov–Ginzburg transform denoted by $\Psi^\dagger$ is defined by

$$\begin{bmatrix} \tilde{z} \\ w \end{bmatrix} = \begin{bmatrix} \Phi_{11} & \Phi_{12} \\ \Phi_{21} & \Phi_{22} \end{bmatrix} \begin{bmatrix} \tilde{w} \\ z \end{bmatrix} = \Psi^\dagger \begin{bmatrix} \tilde{w} \\ z \end{bmatrix}, \quad (13)$$

where

$$\begin{aligned} \Phi_{11} &:= \Psi_{zw} \Psi_w^{-1}, & \Phi_{12} &:= \Psi_z - \Psi_{zw} \Psi_w^{-1} \Psi_{wz}, \\ \Phi_{21} &:= \Psi_w^{-1}, & \Phi_{22} &:= -\Psi_w^{-1} \Psi_{wz}. \end{aligned}$$

Assume that $\Psi^\dagger$ admits the state-space representation

$$\begin{bmatrix} \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} \hat{\mathcal{A}}_\Psi & \hat{\mathcal{B}}_\Psi^{\tilde{w}} & \hat{\mathcal{B}}_\Psi^z \\ \hat{\mathcal{C}}_\Psi^z & \hat{\mathcal{D}}_\Psi^{z\tilde{w}} & \hat{\mathcal{D}}_\Psi^z \\ \hat{\mathcal{C}}_\Psi^w & \hat{\mathcal{D}}_\Psi^{w\tilde{w}} & \hat{\mathcal{D}}_\Psi^{wz} \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \\ z(t) \end{bmatrix}. \quad (14)$$

With $W_\Psi := (\mathcal{D}_\Psi^w)^{-1}$, the transformed blocks are

$$\begin{aligned} \hat{\mathcal{A}}_\Psi &:= \mathcal{A}_\Psi - \mathcal{B}_\Psi^w W_\Psi \mathcal{C}_\Psi^w, & \hat{\mathcal{B}}_\Psi^{\tilde{w}} &:= \mathcal{B}_\Psi^w W_\Psi, \\ \hat{\mathcal{B}}_\Psi^z &:= \mathcal{B}_\Psi^z - \mathcal{B}_\Psi^w W_\Psi \mathcal{D}_\Psi^{wz}, & \hat{\mathcal{C}}_\Psi^z &:= \mathcal{C}_\Psi^z - \mathcal{D}_\Psi^{zw} W_\Psi \mathcal{C}_\Psi^w, \\ \hat{\mathcal{D}}_\Psi^z &:= \mathcal{D}_\Psi^z - \mathcal{D}_\Psi^{zw} W_\Psi \mathcal{D}_\Psi^{wz}, & \hat{\mathcal{D}}_\Psi^{z\tilde{w}} &:= \mathcal{D}_\Psi^{zw} W_\Psi, \\ \hat{\mathcal{C}}_\Psi^w &:= -W_\Psi \mathcal{C}_\Psi^w, & \hat{\mathcal{D}}_\Psi^{w\tilde{w}} &:= W_\Psi, \\ \hat{\mathcal{D}}_\Psi^{wz} &:= -W_\Psi \mathcal{D}_\Psi^{wz}. \end{aligned}$$

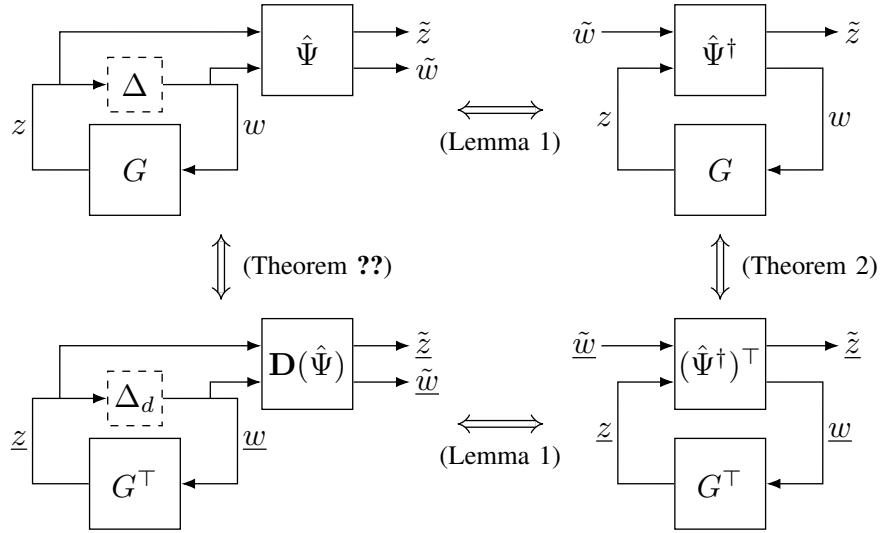


Fig. 1. Commutative diagram for primal and dual dynamic filtering under the Potapov–Ginzburg transform.

Lemma 1. (*Potapov–Ginzburg Transform Equivalence*) Let $\Psi, \mathbf{D}(\Psi) \in H_\infty$ as in Definition 12. Moreover, let G and $G^\top$ be well-posed PIE systems such that

$$\begin{bmatrix} G \star \Psi^\dagger \\ I \end{bmatrix} \in H_\infty, \quad \begin{bmatrix} G^\top \star \mathbf{D}(\Psi)^\dagger \\ I \end{bmatrix} \in H_\infty.$$

Then, there exist invertible bounded causal transformations Γ on the primal side and Γ_d on the dual side such that

- $\Psi \begin{bmatrix} G \\ I \end{bmatrix} w = \begin{bmatrix} G \star \Psi^\dagger \\ I \end{bmatrix} \Gamma w$ for all $w \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w}$.
- $\mathbf{D}(\Psi) \begin{bmatrix} G^\top \\ I \end{bmatrix} \underline{w} = \begin{bmatrix} G^\top \star \mathbf{D}(\Psi)^\dagger \\ I \end{bmatrix} \Gamma_d \underline{w}$ for all $\underline{w} \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z}$.

Proof. See Appendix A.2. ■

V. DUAL IQCS

For brevity we will refer to (7) as the hard IQC

$$\left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_\Psi \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2 \quad (15)$$

with $\Psi = \begin{bmatrix} \Psi & 0 \\ 0 & I \end{bmatrix}$, $w = \begin{bmatrix} w_\Delta \\ w_p \end{bmatrix}$, and $z = \begin{bmatrix} z_\Delta \\ z_p \end{bmatrix}$. The multiplier operator is given by

$$(\mathcal{M}_\Psi \mathbf{u})(t) := \underbrace{\begin{bmatrix} \mathcal{Q}_\Delta & 0 & \mathcal{S}_\Delta & 0 \\ 0 & \mathcal{Q}_p & 0 & \mathcal{S}_p \\ \mathcal{S}_\Delta^* & 0 & \mathcal{R}_\Delta & 0 \\ 0 & \mathcal{S}_p^* & 0 & \mathcal{R}_p \end{bmatrix}}_{\mathcal{V}} \mathbf{u}(t).$$

Definition 14. (*J-spectral factorization, Venkataraman et al. [14]*) $(\hat{\Psi}, J_{\mathbf{n}_z, \mathbf{n}_w})$ is called a *J-spectral factorization* if

$$\begin{aligned} & \left\langle p_T \Psi \begin{bmatrix} z \\ w \end{bmatrix}, p_T \mathcal{M}_\Psi \Psi \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \\ &= \left\langle p_T \hat{\Psi} \begin{bmatrix} z \\ w \end{bmatrix}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \hat{\Psi} \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \end{aligned} \quad (16)$$

with $J_{\mathbf{n}_z, \mathbf{n}_w} = \begin{bmatrix} I_{\mathbf{n}_z} & 0 \\ 0 & -I_{\mathbf{n}_w} \end{bmatrix}$, and $\hat{\Psi}, \mathbf{D}(\hat{\Psi}) \in H_\infty$.

Theorem 3 (Primal–Dual IQC Feasibility). *Let P be the input-output map of a well-posed interconnection as in Definition 7, and let $(\hat{\Psi}, J_{\mathbf{n}_z, \mathbf{n}_w})$ be a J-spectral factorization of the primal hard IQC. Let $\epsilon_p > 0$ and $\epsilon_d > 0$ denote the primal and dual feasibility margins, and set*

$$\epsilon = \min\{\epsilon_p, \epsilon_d\}.$$

Then the primal feasibility condition

$$\begin{aligned} & \left\langle p_T \hat{\Psi} \begin{bmatrix} z \\ w \end{bmatrix}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \hat{\Psi} \begin{bmatrix} z \\ w \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w\|_{\mathbf{L}_{[a,b]}}^2, \\ & z = Pw, \quad w \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w}, \quad T \geq 0, \end{aligned} \quad (17)$$

holds if and only if the dual feasibility condition

$$\begin{aligned} & \left\langle p_T \mathbf{D}(\hat{\Psi}) \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \mathbf{D}(\hat{\Psi}) \begin{bmatrix} \underline{z} \\ \underline{w} \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2, \\ & \underline{z} = P^\top \underline{w}, \quad \underline{w} \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_z}, \quad T \geq 0, \end{aligned} \quad (18)$$

holds for the same $\epsilon > 0$.

Proof. See Appendix A.3. ■

J-spectral factorizations exist for many performance and uncertainty multipliers. Following [13], we collect the relevant multiplier families as classes.

A. Classes of J-Spectral Factorizations for Performance Multipliers

Class 1 (Induced L_2 -gain). *The induced L_2 -gain multiplier class is given by*

$$\mathcal{V}_p = \begin{bmatrix} \gamma^{-1} I_{\mathbf{n}_{z_p}} & 0 \\ 0 & -\gamma I_{\mathbf{n}_{w_p}} \end{bmatrix}, \quad \gamma > 0. \quad (19)$$

Robust performance then guarantees $\sup_{\Delta \in \Delta} \|G \star \Delta\|_{\mathbf{L}_{[a,b]}} \leq \gamma$. A J -spectral factorization is

$$\mathcal{V}_p = \hat{\Psi}_\gamma^* J_{\mathbf{n}_p, \mathbf{n}_w} \hat{\Psi}_\gamma, \quad \hat{\Psi}_\gamma = \begin{bmatrix} \gamma^{-\frac{1}{2}} I_{\mathbf{n}_p} & 0 \\ 0 & \gamma^{\frac{1}{2}} I_{\mathbf{n}_w} \end{bmatrix}. \quad (20)$$

The associated dual factor is

$$\mathbf{D}(\hat{\Psi}_\gamma) = \begin{bmatrix} \gamma^{-\frac{1}{2}} I_{\mathbf{n}_p} & 0 \\ 0 & \gamma^{\frac{1}{2}} I_{\mathbf{n}_w} \end{bmatrix}, \quad J_{\mathbf{n}_p, \mathbf{n}_w}. \quad (21)$$

Class 2 (Passivity). The passivity multiplier class is

$$\mathcal{V}_p = \begin{bmatrix} 0 & I_{\mathbf{n}} \\ I_{\mathbf{n}} & 0 \end{bmatrix}. \quad (22)$$

Robust performance means that $G \star \Delta$ is strictly input passive for all $\Delta \in \Delta$. A J -spectral factorization is

$$\mathcal{V}_p = \hat{\Psi}_{\text{pas}}^* J_{\mathbf{n}, \mathbf{n}} \hat{\Psi}_{\text{pas}}, \quad \hat{\Psi}_{\text{pas}} = \frac{1}{\sqrt{2}} \begin{bmatrix} I_{\mathbf{n}} & I_{\mathbf{n}} \\ I_{\mathbf{n}} & -I_{\mathbf{n}} \end{bmatrix}. \quad (23)$$

The associated dual factor is

$$\mathbf{D}(\hat{\Psi}_{\text{pas}}) = \frac{1}{\sqrt{2}} \begin{bmatrix} -I_{\mathbf{n}} & -I_{\mathbf{n}} \\ -I_{\mathbf{n}} & I_{\mathbf{n}} \end{bmatrix}, \quad J_{\mathbf{n}, \mathbf{n}}. \quad (24)$$

B. Classes of J -Spectral Factorizations for Uncertainty Multipliers

Many finite-dimensional uncertainty IQCs have a strict positive-negative (PN) multiplier structure.

Definition 15 (Strict PN multiplier [8]). Let $\Pi = \Pi^\sim \in \mathbb{RL}_{\infty}^{(n_z + n_w) \times (n_z + n_w)}$, with $\Pi_{11} \in \mathbb{RL}_{\infty}^{n_z \times n_z}$ and $\Pi_{22} \in \mathbb{RL}_{\infty}^{n_w \times n_w}$. The multiplier Π is strict positive-negative if there exists $\epsilon > 0$ such that

$$\Pi_{11}(j\omega) \succeq \epsilon I, \quad \Pi_{22}(j\omega) \preceq -\epsilon I, \quad \forall \omega \in \mathbb{R} \cup \{\infty\}.$$

Moreover, in this finite-dimensional setting, [8] shows that if the primal IQC is a strict PN multiplier, then a J -spectral factorization exists according to Definition 14. This factorization is hard, and all solutions $P = P^\top$ to the associated KYP lemma satisfy $P \geq 0$. Also, [14] shows that the primal IQC is a strict PN multiplier if and only if the corresponding dual multiplier is a strict PN multiplier. Classes 1, 2, 3, 5, 8, and 11 in [13] are strict PN multipliers. Classes 4, 6, 7, 12, 13, 14, and 15 are strict PN after imposing the additional constraints stated in [13].

Class 3 (Real repeated scalar parameters). Consider the finite-dimensional uncertainty class $\Delta_{\text{rr}} := \{\delta I_{n_q} : \delta \in \mathbb{R}, |\delta| \leq \alpha\}$. Define the static matrix scaling

$$\Sigma_\alpha := \begin{bmatrix} \alpha^{-\frac{1}{2}} I_{n_q} & 0 \\ 0 & \alpha^{\frac{1}{2}} I_{n_q} \end{bmatrix} \in \mathbb{R}^{2n_q \times 2n_q}, \quad (25)$$

$$\Psi_{\text{rr}} := \begin{bmatrix} \psi & 0 \\ 0 & \psi \end{bmatrix}.$$

With the convention $w_\Delta = \Delta z_\Delta$, the IQC holds with

$$\Pi_{\text{rr}} = \Sigma_\alpha^* \Psi_{\text{rr}}^* \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix} \Psi_{\text{rr}} \Sigma_\alpha, \quad (26)$$

for all $\Delta \in \Delta_{\text{rr}}$ if $\mathcal{Q} = \mathcal{Q}^* = -\mathcal{R} \in \mathbb{R}^{n_q \times n_q}$, $\mathcal{S} \in \mathbb{R}^{n_q \times n_q}$, and

$$\psi^* \mathcal{Q} \psi \prec 0, \quad \psi^* (\mathcal{S} + \mathcal{S}^*) \psi = 0. \quad (27)$$

Here α is the radius of the real repeated uncertainty block and appears only through the finite-dimensional matrix Σ_α . Since the unscaled dynamic multiplier is strict PN, there exists a hard J -spectral factorization

$$\Psi_{\text{rr}}^* \begin{bmatrix} \mathcal{Q} & \mathcal{S} \\ \mathcal{S}^* & \mathcal{R} \end{bmatrix} \Psi_{\text{rr}} = \hat{\Psi}_{\text{rr}}^* J_{n_q, n_q} \hat{\Psi}_{\text{rr}}, \quad \hat{\Psi}_{\text{rr}}, \mathbf{D}(\hat{\Psi}_{\text{rr}}) \in H_\infty. \quad (28)$$

Thus α remains outside the J -spectral factorization:

$$\Pi_{\text{rr}} = \Sigma_\alpha^* \hat{\Psi}_{\text{rr}}^* J_{n_q, n_q} \hat{\Psi}_{\text{rr}} \Sigma_\alpha. \quad (29)$$

The corresponding unscaled dual multiplier is also strict PN. Hence there exist dual-side parameters $\psi_d, \mathcal{Q}_d, \mathcal{S}_d, \mathcal{R}_d$ such that

$$\mathbf{D}(\hat{\Psi}_{\text{rr}})^* J_{n_q, n_q} \mathbf{D}(\hat{\Psi}_{\text{rr}}) = \begin{bmatrix} \psi_d & 0 \\ 0 & \psi_d \end{bmatrix}^* \begin{bmatrix} \mathcal{Q}_d & \mathcal{S}_d \\ \mathcal{S}_d^* & \mathcal{R}_d \end{bmatrix} \begin{bmatrix} \psi_d & 0 \\ 0 & \psi_d \end{bmatrix}, \quad (30)$$

where $\mathcal{Q}_d = \mathcal{Q}_d^* = -\mathcal{R}_d \in \mathbb{R}^{n_q \times n_q}$, $\mathcal{S}_d \in \mathbb{R}^{n_q \times n_q}$, and

$$\psi_d^* \mathcal{Q}_d \psi_d \prec 0, \quad \psi_d^* (\mathcal{S}_d + \mathcal{S}_d^*) \psi_d = 0. \quad (31)$$

The dual operation is multiplicative for compatible invertible factors:

$$\mathbf{D}(\Psi_1 \Psi_2) = \mathbf{D}(\Psi_1) \mathbf{D}(\Psi_2). \quad (32)$$

Moreover, the static scaling is self-dual,

$$\mathbf{D}(\Sigma_\alpha) = (\hat{\mathcal{J}}^* \Sigma_\alpha^\top \hat{\mathcal{J}})^{-1} = \Sigma_\alpha. \quad (33)$$

Therefore the scaled dual factor is

$$\mathbf{D}(\hat{\Psi}_{\text{rr}} \Sigma_\alpha) = \mathbf{D}(\hat{\Psi}_{\text{rr}}) \Sigma_\alpha, \quad (34)$$

and the scaled dual-side multiplier is obtained by applying the same static matrix scaling outside the dual J -spectral factorization:

$$\Pi_{\text{rr}, d} = \Sigma_\alpha^* \mathbf{D}(\hat{\Psi}_{\text{rr}})^* J_{n_q, n_q} \mathbf{D}(\hat{\Psi}_{\text{rr}}) \Sigma_\alpha. \quad (35)$$

VI. APPLICATION OF DUAL IQC

VII. NUMERICAL EXAMPLES

In this section, we use PIETOOLS [9] to construct the PIE representation and solve the associated LPIs for analysis. As a practical first step, it is generally advisable to synthesize a controller for the nominal system before accounting for uncertainties. In addition, we incorporate performance specifications through weighting filters. This is a standard technique in optimal and robust control synthesis for finite dimensional systems by embedding frequency-dependent performance characteristics into the synthesis.

We begin by considering the boundary control problem for a diffusion–reaction equation, adapted from [7]. Throughout, we assume that the disturbance and control input are measurable, as the introduction of weighting filters gives rise to additional states that depend explicitly on these signals.

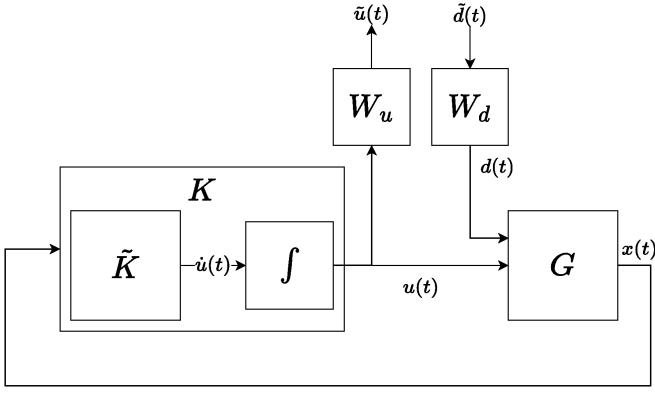


Fig. 2. Block schematic of the weighted diffusion-reaction system with boundary control, where W_u characterizes the inverse frequency-dependent characteristics of the desired control input $u(t)$, and W_d characterizes the frequency content of the disturbance, i.e., $\tilde{u}(t) = W_u u(t)$ and $w(t) = W_d \tilde{w}(t)$.

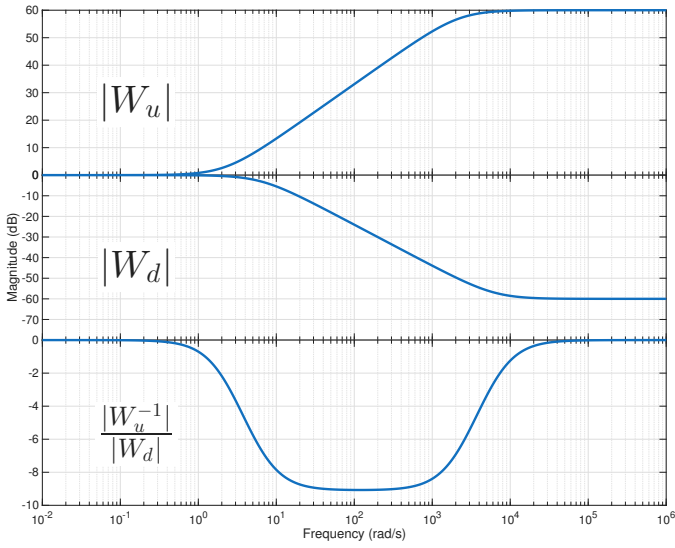


Fig. 3. Bode magnitude plots of the input and disturbance weighting filters, W_u and W_d , together with the combined inverse weighting $W_u^{-1}W_d^{-1}$. These weights are used to encode the desired frequency-dependent characteristics of the control input and disturbance in the weighted boundary control synthesis.

Example 1 (Reaction-Diffusion Equation). *Consider the boundary control of an unstable reaction-diffusion equation:*

$$\begin{aligned}\dot{\mathbf{v}}(t, s) &= 5\mathbf{v}(t, s) + \partial_s^2 \mathbf{v}(t, s) + C_{\tilde{d}} x_{\tilde{d}}(t) + D_{\tilde{d}} \tilde{d}(t), \\ \dot{x}_u(t) &= \dot{u}(t), \\ \dot{x}_{\tilde{u}}(t) &= A_{\tilde{u}} x_{\tilde{u}}(t) + B_{\tilde{u}} x_u(t), \\ \dot{x}_{\tilde{d}}(t) &= A_{\tilde{d}} x_{\tilde{d}}(t) + B_{\tilde{d}} \tilde{d}(t), \\ z(t) &= \tilde{u}(t) = C_{\tilde{u}} x_{\tilde{u}}(t) + D_{\tilde{u}} x_u(t), \\ \mathbf{v}(t, 0) &= 0, \quad \partial_s \mathbf{v}(t, 1) = x_u(t).\end{aligned}$$

Here, the state $x_{\tilde{u}}$ embeds the frequency characteristics of the weighting filter W_u , while $x_{\tilde{d}}$ embeds those of the weighting filter W_d shown in Fig. 3. Moreover, the input u is passed through an integrator so that the control input does not act directly on the boundary. This ensures that $\mathcal{T}_u = 0$, which is required for synthesis. Solving the associated synthesis

conditions yields an $\mathcal{H}_\infty$ gain of $\gamma = 0.9773$, that is,

$$\frac{\|\tilde{u}\|_{L_2}}{\|\tilde{d}\|_{L_2}} < 0.9773.$$

This performance is achieved together with an exponential decay rate of $\alpha = 2$.

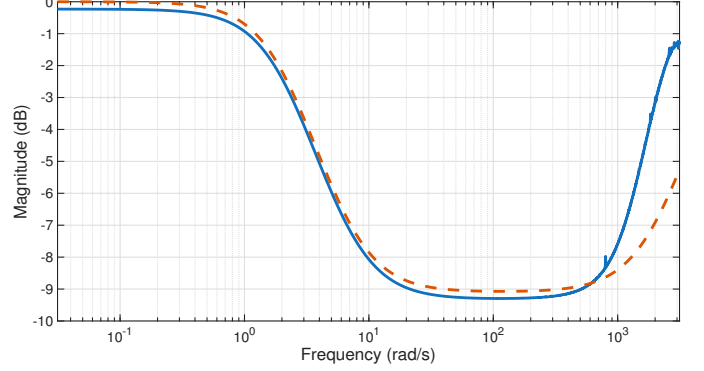


Fig. 4. Empirical frequency-response estimate from the disturbance input to the control input over the range 0.1 rad/s to 10^4 rad/s. — $\frac{|U|}{|W|}$ denotes the estimated transfer magnitude, while - - $\frac{|W_u^{-1}|}{|W_d|}$ denotes the target weighting profile.

To assess whether the weighting specifications are met in practice, we compute an empirical transfer function estimate from the disturbance channel to the control input. For this purpose, we excite the system with a unit-amplitude frequency sweep ranging from 0.1 rad/s to 10^4 rad/s, filtered through the disturbance weighting filter W_d . The resulting signal is applied as the disturbance input, while the corresponding control input is measured. An estimate of the associated frequency response is then obtained from the Fourier transforms of these signals. Comparing this estimate with the desired weighted behavior allows us to verify that the imposed frequency-domain design specifications are achieved.

VIII. CONCLUSION

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APPENDIX A MATHEMATICAL PROOFS

A. Proof of Theorem 1

Suppose that (7) holds. This implies,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta \\ I \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} + \left\langle p_T \begin{bmatrix} P_{p\Delta} \\ 0 \end{bmatrix} w_\Delta, p_T \mathcal{M}_{\mathcal{V}_p} \begin{bmatrix} P_{p\Delta} \\ 0 \end{bmatrix} w_\Delta \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2.$$

Since $\hat{\mathcal{V}}_{11} \succeq 0$, we can then infer,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} P_\Delta w_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2.$$

Then, using Theorem 6 in [11], we find that **robust stability** is guaranteed with respect to performance multiplier $\mathcal{M}_{\hat{\mathcal{V}}}$. Moreover, rewriting (7) as,

$$\left\langle p_T \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_\Delta} \Psi_\Delta \begin{bmatrix} z_\Delta \\ w_\Delta \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} + \left\langle p_T \begin{bmatrix} z_p \\ w_p \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \begin{bmatrix} z_p \\ w_p \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon (\|p_T w_\Delta\|_{\mathbf{L}_{[a,b]}}^2 + \|p_T w_p\|_{\mathbf{L}_{[a,b]}}^2),$$

and given that Ψ_Δ and $\mathcal{M}_{\mathcal{V}_\Delta}$ satisfy the IQC in Definition 10 leads to,

$$\left\langle p_T \begin{bmatrix} z_p \\ w_p \end{bmatrix}, p_T \mathcal{M}_{\mathcal{V}_p} \begin{bmatrix} z_p \\ w_p \end{bmatrix} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon \|p_T w_p\|_{L_2}^2.$$

Thus by Definition 11 **robust performance** is achieved.

B. Proof of Lemma 1

Showing this on operator level might also be nice.

We now record the same equivalence in state-space form. We first show the primal derivation explicitly. To show sufficiency, first augment the primal system with the Potapov–Ginzburg transform of the primal filter, i.e. $G \star \Psi^\dagger$. Substitution of $z = \mathcal{C}\mathbf{x} + \mathcal{D}w$ into (14) gives the algebraic loop

$$w = \Lambda_p \hat{\mathcal{D}}_\Psi^{wz} \mathcal{C}\mathbf{x} + \Lambda_p \hat{\mathcal{C}}_\Psi^w \mathbf{x}_\Psi + \Lambda_p \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \tilde{w}.$$

Assume that this loop is well-posed, and define

$$\Lambda_p := (I - \hat{\mathcal{D}}_\Psi^{wz} \mathcal{D})^{-1}, \quad \Lambda_z := (I - \mathcal{D} \hat{\mathcal{D}}_\Psi^{wz})^{-1}.$$

Solving for w and substituting into the plant state equation gives

$$\mathcal{T} \dot{\mathbf{x}} = \left(\mathcal{A} + \mathcal{B} \Lambda_p \hat{\mathcal{D}}_\Psi^{wz} \mathcal{C} \right) \mathbf{x} + \mathcal{B} \Lambda_p \hat{\mathcal{C}}_\Psi^w \mathbf{x}_\Psi + \mathcal{B} \Lambda_p \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \tilde{w}.$$

Using $\mathcal{D} \Lambda_p = \Lambda_z \mathcal{D}$ and $I + \mathcal{D} \Lambda_p \hat{\mathcal{D}}_\Psi^{wz} = \Lambda_z$, we obtain

$$z = \Lambda_z \mathcal{C}\mathbf{x} + \Lambda_z \mathcal{D} \hat{\mathcal{C}}_\Psi^w \mathbf{x}_\Psi + \Lambda_z \mathcal{D} \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \tilde{w}.$$

Substitution into the equations for $\dot{\mathbf{x}}_\Psi$ and $\tilde{z}$ yields the augmented realization

$$\begin{bmatrix} \mathcal{T} \dot{\mathbf{x}}(t) \\ \mathcal{T}_\Psi \dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} + \mathcal{B} \Lambda_p \hat{\mathcal{D}}_\Psi^{wz} \mathcal{C} & \mathcal{B} \Lambda_p \hat{\mathcal{C}}_\Psi^w & \mathcal{B} \Lambda_p \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \\ \hat{\mathcal{B}}_\Psi^z \Lambda_z \mathcal{C} & \hat{\mathcal{A}}_\Psi + \hat{\mathcal{B}}_\Psi^z \Lambda_z \mathcal{D} \hat{\mathcal{C}}_\Psi^w & \hat{\mathcal{B}}_\Psi^{\tilde{w}} + \hat{\mathcal{B}}_\Psi^z \Lambda_z \mathcal{D} \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \\ \hat{\mathcal{D}}_\Psi^z \Lambda_z \mathcal{C} & \hat{\mathcal{C}}_\Psi^z + \hat{\mathcal{D}}_\Psi^z \Lambda_z \mathcal{D} \hat{\mathcal{C}}_\Psi^w & \hat{\mathcal{D}}_\Psi^{z\tilde{w}} + \hat{\mathcal{D}}_\Psi^z \Lambda_z \mathcal{D} \hat{\mathcal{D}}_\Psi^{w\tilde{w}} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \end{bmatrix} \quad (36)$$

Define also

$$\Lambda_c := (\mathcal{D}_\Psi^{wz} \mathcal{D} + \mathcal{D}_\Psi^w)^{-1}.$$

The invertible congruence transformation used directly on (36) is

$$\begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \mathcal{D}_\Psi^{wz} \mathcal{C} & \mathcal{C}_\Psi^w & \Lambda_c^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ w(t) \end{bmatrix} \quad \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ w(t) \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -\Lambda_c \mathcal{D}_\Psi^{wz} \mathcal{C} & -\Lambda_c \mathcal{C}_\Psi^w & \Lambda_c \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ \tilde{w}(t) \end{bmatrix} \quad (37)$$

The third row of (37) is

$$\tilde{w} = \mathcal{D}_\Psi^{wz} \mathcal{C}\mathbf{x} + \mathcal{C}_\Psi^w \mathbf{x}_\Psi + (\mathcal{D}_\Psi^{wz} \mathcal{D} + \mathcal{D}_\Psi^w) w. \quad (38)$$

Using $z = \mathcal{C}\mathbf{x} + \mathcal{D}w$, this is equivalently

$$\tilde{w} = \mathcal{C}_\Psi^w \mathbf{x}_\Psi + \mathcal{D}_\Psi^{wz} z + \mathcal{D}_\Psi^w w,$$

which is the input row of the direct filter realization.

We now apply (37) directly to (36). The plant row of (36) gives

$$\begin{aligned}\mathcal{T}\dot{\mathbf{x}} &= (\mathcal{A} + \mathcal{B}\Lambda_p\hat{\mathcal{D}}_\Psi^{wz}\mathcal{C})\mathbf{x} + \mathcal{B}\Lambda_p\hat{\mathcal{C}}_\Psi^w\mathbf{x}_\Psi + \mathcal{B}\Lambda_p\hat{\mathcal{D}}_\Psi^{w\tilde{w}}\tilde{w} \\ &= \mathcal{A}\mathbf{x} + \mathcal{B}\left(\Lambda_p\hat{\mathcal{D}}_\Psi^{wz}\mathcal{C}\mathbf{x} + \Lambda_p\hat{\mathcal{C}}_\Psi^w\mathbf{x}_\Psi + \Lambda_p\hat{\mathcal{D}}_\Psi^{w\tilde{w}}\tilde{w}\right) \\ &= \mathcal{A}\mathbf{x} + \mathcal{B}w,\end{aligned}$$

where the last equality uses the algebraic loop solution recovered from the inverse transformation in (37). The filter state row of (36) gives

$$\begin{aligned}\mathcal{T}_\Psi\dot{\mathbf{x}}_\Psi &= \hat{\mathcal{B}}_\Psi^z\Lambda_z\mathcal{C}\mathbf{x} + (\hat{\mathcal{A}}_\Psi + \hat{\mathcal{B}}_\Psi^z\Lambda_z\mathcal{D}\hat{\mathcal{C}}_\Psi^w)\mathbf{x}_\Psi \\ &\quad + (\hat{\mathcal{B}}_\Psi^{\tilde{w}} + \hat{\mathcal{B}}_\Psi^z\Lambda_z\mathcal{D}\hat{\mathcal{D}}_\Psi^{w\tilde{w}})\tilde{w} \\ &= \hat{\mathcal{A}}_\Psi\mathbf{x}_\Psi + \hat{\mathcal{B}}_\Psi^{\tilde{w}}\tilde{w} + \hat{\mathcal{B}}_\Psi^z\left(\Lambda_z\mathcal{C}\mathbf{x} + \Lambda_z\mathcal{D}\hat{\mathcal{C}}_\Psi^w\mathbf{x}_\Psi + \Lambda_z\mathcal{D}\hat{\mathcal{D}}_\Psi^{w\tilde{w}}\tilde{w}\right) \\ &= \hat{\mathcal{A}}_\Psi\mathbf{x}_\Psi + \hat{\mathcal{B}}_\Psi^{\tilde{w}}\tilde{w} + \hat{\mathcal{B}}_\Psi^zz \\ &= (\mathcal{A}_\Psi - \mathcal{B}_\Psi^wW_\Psi\mathcal{C}_\Psi^w)\mathbf{x}_\Psi + \mathcal{B}_\Psi^wW_\Psi(\mathcal{C}_\Psi^w\mathbf{x}_\Psi + \mathcal{D}_\Psi^{wz}z + \mathcal{D}_\Psi^ww) \\ &\quad + (\mathcal{B}_\Psi^z - \mathcal{B}_\Psi^wW_\Psi\mathcal{D}_\Psi^{wz})z \\ &= \mathcal{A}_\Psi\mathbf{x}_\Psi + \mathcal{B}_\Psi^zz + \mathcal{B}_\Psi^ww.\end{aligned}$$

The output row of (36) gives

$$\begin{aligned}\tilde{z} &= \hat{\mathcal{D}}_\Psi^z\Lambda_z\mathcal{C}\mathbf{x} + (\hat{\mathcal{C}}_\Psi^z + \hat{\mathcal{D}}_\Psi^z\Lambda_z\mathcal{D}\hat{\mathcal{C}}_\Psi^w)\mathbf{x}_\Psi \\ &\quad + (\hat{\mathcal{D}}_\Psi^{z\tilde{w}} + \hat{\mathcal{D}}_\Psi^z\Lambda_z\mathcal{D}\hat{\mathcal{D}}_\Psi^{w\tilde{w}})\tilde{w} \\ &= \hat{\mathcal{C}}_\Psi^z\mathbf{x}_\Psi + \hat{\mathcal{D}}_\Psi^{z\tilde{w}}\tilde{w} + \hat{\mathcal{D}}_\Psi^z\left(\Lambda_z\mathcal{C}\mathbf{x} + \Lambda_z\mathcal{D}\hat{\mathcal{C}}_\Psi^w\mathbf{x}_\Psi + \Lambda_z\mathcal{D}\hat{\mathcal{D}}_\Psi^{w\tilde{w}}\tilde{w}\right) \\ &= \hat{\mathcal{C}}_\Psi^z\mathbf{x}_\Psi + \hat{\mathcal{D}}_\Psi^{z\tilde{w}}\tilde{w} + \hat{\mathcal{D}}_\Psi^zz \\ &= (\mathcal{C}_\Psi^z - \mathcal{D}_\Psi^{zw}W_\Psi\mathcal{C}_\Psi^w)\mathbf{x}_\Psi + \mathcal{D}_\Psi^{zw}W_\Psi(\mathcal{C}_\Psi^w\mathbf{x}_\Psi + \mathcal{D}_\Psi^{wz}z + \mathcal{D}_\Psi^ww) \\ &\quad + (\mathcal{D}_\Psi^z - \mathcal{D}_\Psi^{zw}W_\Psi\mathcal{D}_\Psi^{wz})z \\ &= \mathcal{C}_\Psi^z\mathbf{x}_\Psi + \mathcal{D}_\Psi^zz + \mathcal{D}_\Psi^{zw}w.\end{aligned}$$

Substituting $z = \mathcal{C}\mathbf{x} + \mathcal{D}w$ in these recovered rows gives

$$\begin{bmatrix} \mathcal{T}\dot{\mathbf{x}}(t) \\ \mathcal{T}_\Psi\dot{\mathbf{x}}_\Psi(t) \\ \tilde{z}(t) \\ \tilde{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A} & 0 & \mathcal{B} \\ \mathcal{B}_\Psi^z\mathcal{C} & \mathcal{A}_\Psi & \mathcal{B}_\Psi^z\mathcal{D} + \mathcal{B}_\Psi^w \\ \mathcal{D}_\Psi^z\mathcal{C} & \mathcal{C}_\Psi^z & \mathcal{D}_\Psi^z\mathcal{D} + \mathcal{D}_\Psi^{zw} \\ \mathcal{D}_\Psi^{wz}\mathcal{C} & \mathcal{C}_\Psi^w & \mathcal{D}_\Psi^{wz}\mathcal{D} + \mathcal{D}_\Psi^w \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \mathbf{x}_\Psi(t) \\ w(t) \end{bmatrix}. \quad (39)$$

The realization in (39) is the direct augmentation $\Psi \begin{bmatrix} G \\ I \end{bmatrix} w$. For necessity, applying the inverse of (37) to (39) recovers (36).

Dual side. The dual calculation has the same structure. The inverse relation $\mathbf{D}(\Psi)^{-1} = \hat{\mathcal{J}}^*\Psi^\top\hat{\mathcal{J}}$ has realization

$$\begin{bmatrix} \mathcal{T}_\Psi^*\dot{\mathbf{x}}_\Psi(t) \\ \underline{z}(t) \\ \underline{w}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi^* & -\mathcal{C}_\Psi^{w*} & \mathcal{C}_\Psi^{z*} \\ -\mathcal{B}_\Psi^{w*} & \mathcal{D}_\Psi^{w*} & -\mathcal{D}_\Psi^{zw*} \\ \mathcal{B}_\Psi^{z*} & -\mathcal{D}_\Psi^{wz*} & \mathcal{D}_\Psi^{z*} \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix}. \quad (40)$$

Solving the second block row for $\underline{\tilde{z}}$ and substituting in the first and third block rows gives the Potapov–Ginzburg realization of the dual filter:

$$\begin{bmatrix} \mathcal{T}_\Psi^*\dot{\mathbf{x}}_\Psi(t) \\ \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix} = \begin{bmatrix} (\hat{\mathcal{A}}_\Psi)^* & (\hat{\mathcal{C}}_\Psi^z)^* & (\hat{\mathcal{C}}_\Psi^w)^* \\ (\hat{\mathcal{B}}_\Psi^{\tilde{w}})^* & (\hat{\mathcal{D}}_\Psi^{z\tilde{w}})^* & (\hat{\mathcal{D}}_\Psi^{w\tilde{w}})^* \\ (\hat{\mathcal{B}}_\Psi^z)^* & (\hat{\mathcal{D}}_\Psi^z)^* & (\hat{\mathcal{D}}_\Psi^{wz})^* \end{bmatrix} \begin{bmatrix} \mathbf{x}_\Psi(t) \\ \underline{\tilde{w}}(t) \\ \underline{\tilde{z}}(t) \end{bmatrix}. \quad (41)$$

Augmenting (41) with $G^\top$ and using $\underline{z} = \mathcal{B}^*\mathbf{x} + \mathcal{D}^*\underline{w}$ gives the dual algebraic loop

$$\underline{w} = (\hat{\mathcal{D}}_\Psi^{wz})^*\mathcal{B}^*\mathbf{x} + (\hat{\mathcal{B}}_\Psi^z)^*\mathbf{x}_\Psi + (\hat{\mathcal{D}}_\Psi^z)^*\underline{\tilde{w}} + (\hat{\mathcal{D}}_\Psi^{wz})^*\mathcal{D}^*\underline{w}.$$

Define

$$\Lambda_z^* := (I - (\hat{\mathcal{D}}_\Psi^{wz})^*\mathcal{D}^*)^{-1}, \quad \Lambda_p^* := (I - \mathcal{D}^*(\hat{\mathcal{D}}_\Psi^{wz})^*)^{-1}.$$

Then

$$\underline{w} = \Lambda_z^*(\hat{\mathcal{D}}_\Psi^{wz})^*\mathcal{B}^*\mathbf{x} + \Lambda_z^*(\hat{\mathcal{B}}_\Psi^z)^*\mathbf{x}_\Psi + \Lambda_z^*(\hat{\mathcal{D}}_\Psi^z)^*\underline{\tilde{w}},$$

and, using $\mathcal{D}^* \Lambda_z^* = \Lambda_p^* \mathcal{D}^*$ and $I + \mathcal{D}^* \Lambda_z^* (\hat{\mathcal{D}}_\Psi^{wz})^* = \Lambda_p^*$,

$$\underline{z} = \Lambda_p^* \mathcal{B}^* \underline{x} + \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{B}}_\Psi^z)^* \underline{x}_\Psi + \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{D}}_\Psi^z)^* \underline{w}.$$

Substitution into the plant row and the first two rows of (41) yields

$$\begin{bmatrix} \mathcal{T}^* \dot{\underline{x}}(t) \\ \mathcal{T}_\Psi^* \dot{\underline{x}}_\Psi(t) \\ \underline{\tilde{z}}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* + \mathcal{C}^* \Lambda_z^* (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{B}^* & \mathcal{C}^* \Lambda_z^* (\hat{\mathcal{B}}_\Psi^z)^* & \mathcal{C}^* \Lambda_z^* (\hat{\mathcal{D}}_\Psi^z)^* \\ (\hat{\mathcal{C}}_\Psi^w)^* \Lambda_p^* \mathcal{B}^* & (\hat{\mathcal{A}}_\Psi)^* + (\hat{\mathcal{C}}_\Psi^w)^* \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{B}}_\Psi^z)^* & (\hat{\mathcal{C}}_\Psi^z)^* + (\hat{\mathcal{C}}_\Psi^w)^* \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{D}}_\Psi^z)^* \\ (\hat{\mathcal{D}}_\Psi^{w\tilde{w}})^* \Lambda_p^* \mathcal{B}^* & (\hat{\mathcal{B}}_\Psi^{\tilde{w}})^* + (\hat{\mathcal{D}}_\Psi^{w\tilde{w}})^* \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{B}}_\Psi^z)^* & (\hat{\mathcal{D}}_\Psi^{z\tilde{w}})^* + (\hat{\mathcal{D}}_\Psi^{w\tilde{w}})^* \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{D}}_\Psi^z)^* \end{bmatrix} \begin{bmatrix} \underline{x}(t) \\ \underline{x}_\Psi(t) \\ \underline{\tilde{w}}(t) \end{bmatrix}. \quad (42)$$

The loop solution above gives the invertible input transformation

$$\begin{bmatrix} \underline{x}(t) \\ \underline{x}_\Psi(t) \\ \underline{w}(t) \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ \Lambda_z^* (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{B}^* & \Lambda_z^* (\hat{\mathcal{B}}_\Psi^z)^* & \Lambda_z^* (\hat{\mathcal{D}}_\Psi^z)^* \end{bmatrix} \begin{bmatrix} \underline{x}(t) \\ \underline{x}_\Psi(t) \\ \underline{\tilde{w}}(t) \end{bmatrix}, \quad (43)$$

with inverse

$$\begin{bmatrix} \underline{x}(t) \\ \underline{x}_\Psi(t) \\ \underline{\tilde{w}}(t) \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ -((\hat{\mathcal{D}}_\Psi^z)^*)^{-1} (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{B}^* & -((\hat{\mathcal{D}}_\Psi^z)^*)^{-1} (\hat{\mathcal{B}}_\Psi^z)^* & ((\hat{\mathcal{D}}_\Psi^z)^*)^{-1} (I - (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{D}^*) \end{bmatrix} \begin{bmatrix} \underline{x}(t) \\ \underline{x}_\Psi(t) \\ \underline{w}(t) \end{bmatrix}. \quad (44)$$

The third block row of (44) gives the PIE realization of the dual input transformation $\Gamma_d \underline{w} = \underline{\tilde{w}}$. We now show explicitly that applying (44) to (42) gives the direct filtered augmentation.

Input row. The third row of (44) is

$$(\hat{\mathcal{D}}_\Psi^z)^* \underline{\tilde{w}} = \underline{w} - (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{B}^* \underline{x} - (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{D}^* \underline{w} - (\hat{\mathcal{B}}_\Psi^z)^* \underline{x}_\Psi.$$

Using $\underline{z} = \mathcal{B}^* \underline{x} + \mathcal{D}^* \underline{w}$ and rearranging gives

$$\underline{w} = (\hat{\mathcal{B}}_\Psi^z)^* \underline{x}_\Psi + (\hat{\mathcal{D}}_\Psi^z)^* \underline{\tilde{w}} + (\hat{\mathcal{D}}_\Psi^{wz})^* \underline{z}, \quad (45)$$

which is the recovered algebraic row of (41).

Plant row. The first row of (42) contains the loop-solved input. By (45), this input reduces to $\underline{w}$:

$$\begin{aligned} & \Lambda_z^* (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{B}^* \underline{x} + \Lambda_z^* (\hat{\mathcal{B}}_\Psi^z)^* \underline{x}_\Psi + \Lambda_z^* (\hat{\mathcal{D}}_\Psi^z)^* \underline{\tilde{w}} \\ &= \Lambda_z^* (\underline{w} - (\hat{\mathcal{D}}_\Psi^{wz})^* \mathcal{D}^* \underline{w}) = \underline{w}. \end{aligned}$$

The corresponding loop-solved plant output reduces to $\underline{z}$:

$$\begin{aligned} & \Lambda_p^* \mathcal{B}^* \underline{x} + \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{B}}_\Psi^z)^* \underline{x}_\Psi + \Lambda_p^* \mathcal{D}^* (\hat{\mathcal{D}}_\Psi^z)^* \underline{\tilde{w}} \\ &= \Lambda_p^* (\mathcal{B}^* \underline{x} + \mathcal{D}^* \underline{w} - \mathcal{D}^* (\hat{\mathcal{D}}_\Psi^{wz})^* \underline{z}) = \underline{z}. \end{aligned}$$

Therefore the transformed plant row is

$$\mathcal{T}^* \dot{\underline{x}} = \mathcal{A}^* \underline{x} + \mathcal{C}^* \underline{w}.$$

Filter rows. The remaining rows are obtained by solving the algebraic part of (40). Define

$$\mathcal{D}_\Psi := \begin{bmatrix} \mathcal{D}_\Psi^{w*} & -\mathcal{D}_\Psi^{z*} \\ -\mathcal{D}_\Psi^{wz*} & \mathcal{D}_\Psi^{z*} \end{bmatrix}.$$

The two algebraic rows of (40) are

$$\mathcal{D}_\Psi \begin{bmatrix} \underline{\tilde{z}} \\ \underline{\tilde{w}} \end{bmatrix} = \begin{bmatrix} \mathcal{B}_\Psi^{w*} \\ -\mathcal{B}_\Psi^{z*} \end{bmatrix} \underline{x}_\Psi + \begin{bmatrix} \underline{\tilde{z}} \\ \underline{w} \end{bmatrix}.$$

Since $\mathcal{D}_\Psi$ is invertible, solving these rows and substituting in the state row of (40) gives

$$\begin{bmatrix} \mathcal{T}_\Psi^* \dot{\underline{x}}_\Psi(t) \\ \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix} = \begin{bmatrix} \mathcal{A}_\Psi^* + [-\mathcal{C}_\Psi^{w*} & \mathcal{C}_\Psi^{z*}] \mathcal{D}_\Psi^{-1} \begin{bmatrix} \mathcal{B}_\Psi^{w*} \\ -\mathcal{B}_\Psi^{z*} \end{bmatrix} & [-\mathcal{C}_\Psi^{w*} & \mathcal{C}_\Psi^{z*}] \mathcal{D}_\Psi^{-1} \\ \mathcal{D}_\Psi^{-1} \begin{bmatrix} \mathcal{B}_\Psi^{w*} \\ -\mathcal{B}_\Psi^{z*} \end{bmatrix} & \mathcal{D}_\Psi^{-1} \end{bmatrix} \begin{bmatrix} \underline{x}_\Psi(t) \\ \underline{\tilde{z}}(t) \\ \underline{w}(t) \end{bmatrix}. \quad (46)$$

Finally substituting $\underline{z} = \mathcal{B}^* \underline{x} + \mathcal{D}^* \underline{w}$ into (46), and collecting it with $\mathcal{T}^* \dot{\underline{x}} = \mathcal{A}^* \underline{x} + \mathcal{C}^* \underline{w}$, gives exactly

$$\begin{bmatrix} \mathcal{T}^* \dot{\underline{x}}(t) \\ \mathcal{T}_\Psi^* \dot{\underline{x}}_\Psi(t) \\ \begin{bmatrix} \underline{\tilde{z}}(t) \\ \underline{\tilde{w}}(t) \end{bmatrix} \end{bmatrix} = \begin{bmatrix} \mathcal{A}^* & 0 & \mathcal{C}^* \\ [-\mathcal{C}_\Psi^{w*} & \mathcal{C}_\Psi^{z*}] \mathcal{D}_\Psi^{-1} \begin{bmatrix} \mathcal{B}^* \\ 0 \end{bmatrix} & \mathcal{A}_\Psi^* + [-\mathcal{C}_\Psi^{w*} & \mathcal{C}_\Psi^{z*}] \mathcal{D}_\Psi^{-1} \begin{bmatrix} \mathcal{B}_\Psi^{w*} \\ -\mathcal{B}_\Psi^{z*} \end{bmatrix} & [-\mathcal{C}_\Psi^{w*} & \mathcal{C}_\Psi^{z*}] \mathcal{D}_\Psi^{-1} \begin{bmatrix} \mathcal{D}^* \\ I \end{bmatrix} \\ \mathcal{D}_\Psi^{-1} \begin{bmatrix} \mathcal{B}^* \\ 0 \end{bmatrix} & \mathcal{D}_\Psi^{-1} \begin{bmatrix} \mathcal{B}_\Psi^{w*} \\ -\mathcal{B}_\Psi^{z*} \end{bmatrix} & \mathcal{D}_\Psi^{-1} \begin{bmatrix} \mathcal{D}^* \\ I \end{bmatrix} \end{bmatrix} \begin{bmatrix} \underline{x}(t) \\ \underline{x}_\Psi(t) \\ \underline{w}(t) \end{bmatrix}. \quad (47)$$

This realization represents $\mathbf{D}(\Psi) \begin{bmatrix} G^\top \\ I \end{bmatrix} \underline{w}$.

C. Proof of Theorem 3

Assume that the primal statement of the theorem holds. Then there exists a sufficiently small $\epsilon_p > 0$ such that

$$\left\langle p_T \hat{\Psi} \begin{bmatrix} P \\ I \end{bmatrix} w, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \hat{\Psi} \begin{bmatrix} P \\ I \end{bmatrix} w \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon_p \|p_T w\|_{\mathbf{L}_{[a,b]}}^2 \quad (48)$$

for all $w \in \mathbf{L}_{e,[a,b]}^{\mathbf{n}_w}$ and $T \geq 0$. By lemma 1, the system $\hat{\Psi} \begin{bmatrix} P \\ I \end{bmatrix}$ has a realization of the form

$$\left\langle p_T \begin{bmatrix} P \star \hat{\Psi}^\dagger \\ I \end{bmatrix} \Gamma w, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \begin{bmatrix} P \star \hat{\Psi}^\dagger \\ I \end{bmatrix} \Gamma w \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon_p \|p_T w\|_{\mathbf{L}_{[a,b]}}^2 \quad (49)$$

substituting $\tilde{w} = \Gamma w$, $\Gamma^{-1} \tilde{w} = w$, gives

$$\left\langle p_T \begin{bmatrix} P \star \hat{\Psi}^\dagger \\ I \end{bmatrix} \tilde{w}, p_T J_{\mathbf{n}_z, \mathbf{n}_w} \begin{bmatrix} P \star \hat{\Psi}^\dagger \\ I \end{bmatrix} \tilde{w} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon_p \|p_T \Gamma^{-1} \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2 \quad (50)$$

Since Γ is causal and bounded, $\|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}} \leq \|\Gamma\|_{\mathbf{L}_{[a,b]}} \|p_T \Gamma^{-1} \tilde{w}\|_{\mathbf{L}_{[a,b]}}$, and therefore

$$\|p_T (P \star \hat{\Psi}^\dagger) \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2 \leq \left(1 - \frac{\epsilon_p}{\|\Gamma\|_{\mathbf{L}_{[a,b]}}^2}\right) \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2. \quad (51)$$

Then by Theorem 2, and $(P \star \hat{\Psi}^\dagger)^\top = P^\top \star \mathbf{D}(\hat{\Psi})^\dagger$, we find

$$\|p_T (P^\top \star \mathbf{D}(\hat{\Psi})^\dagger) \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2 \leq \left(1 - \frac{\epsilon_p}{\|\Gamma\|_{\mathbf{L}_{[a,b]}}^2}\right) \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2 \quad (52)$$

recovering the dual statement of the theorem

$$\left\langle p_T \begin{bmatrix} P^\top \star \mathbf{D}(\hat{\Psi})^\dagger \\ I \end{bmatrix} \tilde{w}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \begin{bmatrix} P^\top \star \mathbf{D}(\hat{\Psi})^\dagger \\ I \end{bmatrix} \tilde{w} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\frac{\epsilon_p}{\|\Gamma\|_{\mathbf{L}_{[a,b]}}^2} \|p_T \tilde{w}\|_{\mathbf{L}_{[a,b]}}^2 \quad (53)$$

then $\underline{w} = \Gamma_d \underline{w}$, $\Gamma_d^{-1} \underline{w} = \underline{w}$ gives

$$\left\langle p_T \begin{bmatrix} P^\top \star \mathbf{D}(\hat{\Psi})^\dagger \\ I \end{bmatrix} \Gamma_d \underline{w}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \begin{bmatrix} P^\top \star \mathbf{D}(\hat{\Psi})^\dagger \\ I \end{bmatrix} \Gamma_d \underline{w} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\frac{\epsilon_p}{\|\Gamma\|_{\mathbf{L}_{[a,b]}}^2} \|p_T \Gamma_d \underline{w}\|_{\mathbf{L}_{[a,b]}}^2 \quad (54)$$

Since Γ_d^{-1} is causal and bounded, $\|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}} \leq \|\Gamma_d^{-1}\|_{\mathbf{L}_{[a,b]}} \|p_T \Gamma_d \underline{w}\|_{\mathbf{L}_{[a,b]}}$, and this gives the dual statement

$$\left\langle p_T \mathbf{D}(\hat{\Psi}) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w}, p_T J_{\mathbf{n}_w, \mathbf{n}_z} \mathbf{D}(\hat{\Psi}) \begin{bmatrix} P^\top \\ I \end{bmatrix} \underline{w} \right\rangle_{\mathbf{L}_{[a,b]}} \leq -\epsilon_d \|p_T \underline{w}\|_{\mathbf{L}_{[a,b]}}^2 \quad (55)$$

where

$$\epsilon_d := \frac{\epsilon_p}{\|\Gamma\|_{\mathbf{L}_{[a,b]}}^2 \|\Gamma_d^{-1}\|_{\mathbf{L}_{[a,b]}}^2}. \quad (56)$$

Conversely, starting from the dual statement and reversing the same argument gives the corresponding primal margin

$$\epsilon_p := \frac{\epsilon_d}{\|\Gamma_d\|_{\mathbf{L}_{[a,b]}}^2 \|\Gamma^{-1}\|_{\mathbf{L}_{[a,b]}}^2}. \quad (57)$$

Thus, in either direction, the common margin in the theorem is obtained by taking

$$\epsilon = \min\{\epsilon_p, \epsilon_d\}. \quad (58)$$